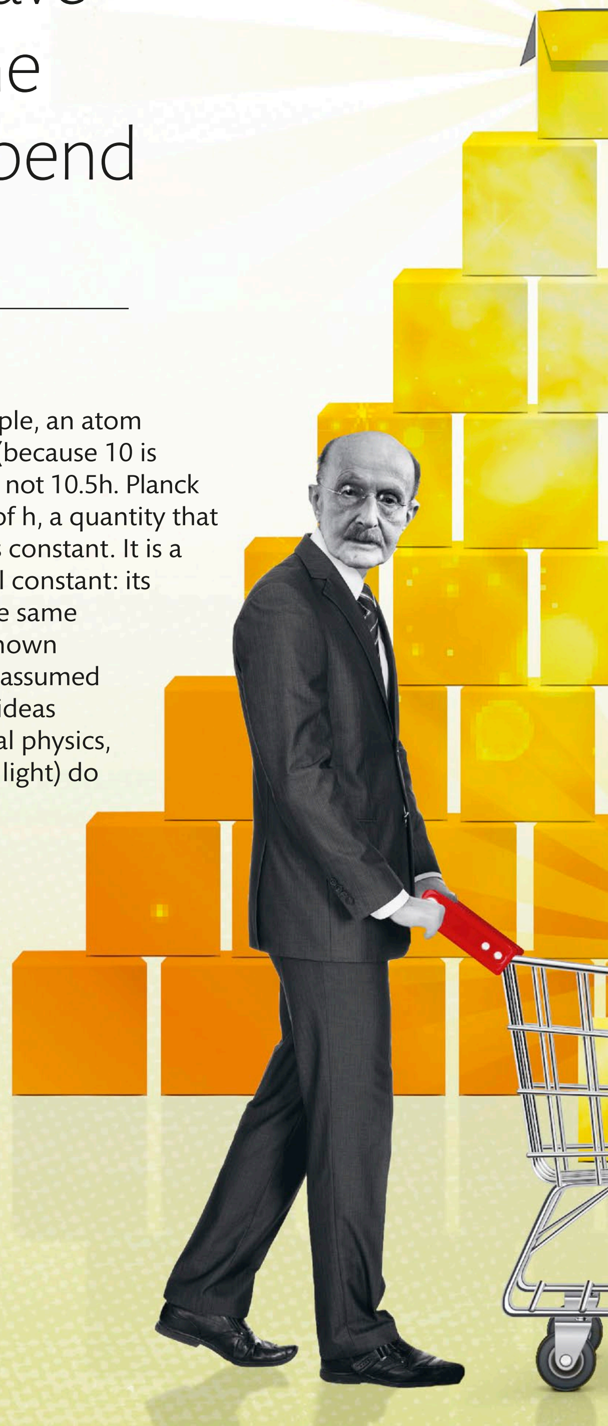
“The **laws of Physics** have **no consideration** for the **human senses**; they depend on **the facts** ...”

Max Planck, 1931

Planck investigated how the intensity of the electromagnetic radiation emitted by a black body depended on the body’s temperature and the frequency of the radiation (the color of the light). At the time, physicists assumed that the sources of radiation were atoms that could oscillate (vibrate) continuously at any frequency. But by 1899, Planck had noted that atoms could only vibrate at frequencies that were whole-number multiples of a base frequency that he named “h.” For example, an atom could vibrate at 10h (because 10 is a whole number) but not 10.5h. Planck calculated the value of h, a quantity that is now called Planck’s constant. It is a fundamental physical constant: its numerical value is the same everywhere in the known Universe. Planck also assumed that, contrary to the ideas postulated by classical physics, photons (particles of light) do



PAUL **DIRAC**

English physicist Paul Dirac is best known for his Dirac equation, which predicted the existence of antimatter particles such as the positron.

As a postgraduate student, Dirac (1902–1984) read Werner Heisenberg’s paper on matrix mechanics describing how particles jump from one quantum state to another. He could see parallels with parts of the classical (prequantum) theory of particle motion. He worked out a way to understand classical systems on a quantum level and created quantum field theory. His Dirac equation predicted “antimatter” or “positrons”—particles with identical properties to particles of matter but with the opposite electrical charge. In 1932, Dirac was appointed Lucasian Professor of Mathematics at Cambridge University and, with Erwin Schrödinger, was awarded the Nobel Prize in Physics in 1933.